

Spheres of Linear Transformations and Matrices

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Journal of Undergraduate Mathematics
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Preface

Many characterizations of circles have been given synthetically and analytically. A vector approach to this study is quite powerful, and can be generalized to the space of linear transformations on a unitary space. We shall study a few samples, and suggest some problems for undergraduate research in mathematics.

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1 Unitary Spaces

In what follows, we consider only vector spaces over the field of real or complex numbers. Vectors will be denoted by Greek letters ($\alpha, \beta, \gamma, \dots$), and scalars by Latin letters ($a, b, c, \dots$).

1.1 Inner Product

Let V be a vector space over a field of complex numbers $\mathbb{C}$. Let f be a function of $V \times V$ into $\mathbb{C}$, such that:

$$f(\eta, \xi) = \overline{f(\xi, \eta)} \quad (1)$$

where $\overline{f(\xi, \eta)}$ is the complex conjugate of $f(\xi, \eta)$,

$$f(\xi, \xi) \geq 0, \text{ and } f(\xi, \xi) = 0 \text{ if and only if } \xi = \vec{0} \quad (2)$$

$$f(a_1\xi_1 + a_2\xi_2, \eta) = a_1f(\xi_1, \eta) + a_2f(\xi_2, \eta) \quad (3)$$

for all ξ_1, ξ_2, η in V and a_1, a_2 in $\mathbb{C}$. f is called an *inner product* on V . For convenience, we write (ξ, η) for $f(\xi, \eta)$. The *norm* of a vector ξ will be defined by $\|\xi\| = (\xi, \xi)^{1/2}$. In the case where V is over the field of real numbers, the inner product is commutative, i.e. $(\xi, \eta) = (\eta, \xi)$.

A vector space V over the field of real or complex numbers is called *unitary* if an inner product is defined for V . Such a space is also called an *inner product* space.

A unitary space over the field of complex numbers will be denoted by $\mathbb{E}$, and a unitary space over the field of real numbers will be denoted by $\mathbb{R}$. When the space is finite-dimensional with dimension n , we denote it by $\mathbb{E}_n$ if it is complex, and by $\mathbb{R}_n$ if it is real.

1.2 Linear Transformations

Let A be a linear transformation on $\mathbb{E}_n$. Then the adjoint of A is denoted by A^* , and is defined by

$$(A\xi, \eta) = (\xi, A^*\eta), \quad \xi, \eta \in \mathbb{E}_n$$

If $A^* = A$, then A is called a *Hermitian* or a *self-adjoint* transformation. If $A^*A = AA^*$, then A is called a *normal* transformation. If $A^*A = AA^* = I$, then A is a *unitary* transformation. A Hermitian transformation A is called *positive* (*positive definite*) if for every $\xi \neq 0, \xi \in \mathbb{E}_n$, we have $(A\xi, \xi) > 0$. A Hermitian transformation A is called *non-negative* (*positive semidefinite*) if for every $\xi \in \mathbb{E}_n$, we have $(A\xi, \xi) \geq 0$.

Now we shall review a few theorems without proof. Proper values of a positive (non-negative) transformation are positive (non-negative). Let $m_1, \dots, m_n$ be proper values of a Hermitian transformation A . Then there exists an orthonormal basis $\{\alpha_1, \dots, \alpha_n\}$ such that

$$A\alpha_i = m_i\alpha_i, \quad i = 1, \dots, n$$

Whenever necessary, we shall state other theorems [1, 2].

Theorem 1.1 (Fischer): Let A be a Hermitian transformation on $\mathbb{E}_n$, with proper values $m_1 \geq \dots \geq m_n$. Then

$$m_1 = \sup_{\|\xi\|=1} (A\xi, \xi) \quad (4)$$

and

$$m_n = \inf_{\|\xi\|=1} (A\xi, \xi) \quad (5)$$

Proof: Let $\{\alpha_1, \dots, \alpha_n\}$ be an orthonormal set of proper vectors of A such that $A\alpha_i = m_i\alpha_i, i = 1, \dots, n$. Then for any $\xi \in \mathbb{E}_n$, we have

$$\xi = (\xi, \alpha_1)\alpha_1 + \dots + (\xi, \alpha_n)\alpha_n$$

and

$$A\xi = m_1(\xi, \alpha_1)\alpha_1 + \dots + m_n(\xi, \alpha_n)\alpha_n$$

Therefore

$$(A\xi, \xi) = m_1|(\xi, \alpha_1)|^2 + \dots + |(\xi, \alpha_n)|^2$$

To prove (1), we observe that

$$(A\xi, \xi) \leq m_1 [|(\xi, \alpha_1)|^2 + \dots + |(\xi, \alpha_n)|^2]$$

Now if we let $\|\xi\| = 1$, then we have

$$1 = \|\xi\|^2 = |(\xi, \alpha_1)|^2 + \dots + |(\xi, \alpha_n)|^2$$

Therefore, $(A\xi, \xi) \leq m_1$. If we let $\xi = \alpha_1$, we obtain $(A\xi, \xi) = (A\alpha_1, \alpha_1) = m_1$. Thus $\sup_{\|\xi\|=1} (A\xi, \xi) = m_1$. Now to prove (2), one notes that

$$(A\xi, \xi) \geq m_n [|(\xi, \alpha_1)|^2 + \dots + |(\xi, \alpha_n)|^2]$$

If we let $\|\xi\| = 1$, we obtain $(A\xi, \xi) \geq m_n$. Again for $\xi = \alpha_n$, we have

$$(A\xi, \xi) = (A\alpha_n, \alpha_n) = m_n$$

Consequently, $\inf_{\|\xi\|=1} (A\xi, \xi) = m_n$.

Many generalizations of this result may be found in the literature [1]. Since we do not use them, we shall not pursue that direction.

Theorem 1.2: Let A be a linear transformation on $\mathbb{E}_n$. Then A^*A and AA^* are both non-negative, and have the same proper values.

We omit the proof.

Definition 1.1 (Square Roots): We shall define non-negative square roots only for non-negative transformations. Let A be a non-negative transformation on $\mathbb{E}_n$, with proper values $h_1 \geq \cdots \geq h_n$. Let $\{\alpha_1, \dots, \alpha_n\}$ be an orthonormal basis in $\mathbb{E}_n$ such that $A\alpha_i = h_i\alpha_i$, $i = 1, \dots, n$. Then the matrix of A with respect to this basis is

$$\begin{pmatrix} h_1 & & 0 \\ & \ddots & \\ 0 & & h_n \end{pmatrix}$$

which is in diagonal form. We define the linear transformation B on $\mathbb{E}_n$ whose matrix with respect to $\{\alpha_1, \dots, \alpha_n\}$ is

$$\begin{pmatrix} \sqrt{h_1} & & 0 \\ & \ddots & \\ 0 & & \sqrt{h_n} \end{pmatrix}$$

Then B is called the *non-negative square root* of A , and will be denoted by $\sqrt{A}$. It is clear that $\sqrt{A}$ is non-negative and has the same proper vectors as A .

Definition 1.2 (Singular Values): Let A be a linear transformation on $\mathbb{E}_n$, and let $a_1, \dots, a_n$ be proper values of A^*A . Then the set $\{\sqrt{a_1}, \dots, \sqrt{a_n}\}$ is called the set of *absolute singular values* of A . Note that we have only considered positive square roots. In what follows, whenever we say “singular values”, we mean “absolute singular values”.

We may easily show that the absolute singular values of A are the same as the proper values of $\sqrt{A^*A}$ or the proper values of $\sqrt{AA^*}$. We leave the proof as an exercise.

Definition 1.3 (The Hilbert Norm): Let A be a linear transformation on $\mathbb{E}_n$. Then the *Hilbert norm* of A is defined by

$$\|A\|_H = \sup_{\|\xi\|=1} \|A\xi\|$$

One can express this norm in terms of singular values of A . Let a be the singular value of A such that a is maximal. We observe that a^2 is a proper value of A^*A . By Theorem 1.1, equation (4), we have $\sup_{\|\xi\|=1} (A^*A\xi, \xi) = a^2$, but $(A^*A\xi, \xi) = (A\xi, A\xi) = \|A\xi\|^2$. Therefore, $\|A\|_H^2 = \sup_{\|\xi\|=1} \|A\xi\|^2 = a^2$. This implies that $\|A\|_H = a$.

Definition 1.4 (Frobenius Inner Product): Let A and B be linear transformations on $\mathbb{E}_n$. Let $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$ be an orthonormal basis on $\mathbb{E}_n$. Let (a_{ij}) and (b_{ij}) respectively be the matrices of A and B with respect to $\mathcal{B}$. Then we define the *Frobenius inner product* of A and B by

$$(A, B)_F = \sum_{i,j=1}^n a_{ij} \bar{b}_{ij}$$

For example, let

$$[A] = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}$$

and

$$[B] = \begin{pmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & & \vdots \\ b_{n1} & \cdots & b_{nn} \end{pmatrix}$$

Then $(A, B)_F = a_{11}\bar{b}_{11} + a_{12}\bar{b}_{12} + \cdots + a_{nn}\bar{b}_{nn}$. This inner product introduces the Frobenius norm for A , i.e.,

$$\|A\|_F^2 = (A, A)_F = \sum_{i,j=1}^n (a_{ij})^2$$

The next theorem will show that this inner product is well-defined; i.e., it is independent of the choice of the basis.

Theorem 1.3: Let A and B be linear transformations on $\mathbb{E}_n$. Then

$$(A, B)_F = \text{trace } AB^*$$

Theorem 1.4: Let A be a linear transformation on $\mathbb{E}_n$. Let $a_1, \dots, a_n$ be the singular values of A . Then

$$\|A\|_F = \left(\sum_{i=1}^n a_i^2 \right)^{1/2}$$

2 Spheres of Transformations

In what follows, all transformations are linear on $\mathbb{E}_n$ or $\mathbb{R}_n$.

Definition 2.1 (Adjoint Products): Let A and B be linear transformations. We define A^*B to be the *adjoint product* or **-product* of A and B .

This product does not have all the properties of an inner product. But one may consider analogues of geometrical problems for this product.

Lemma 2.1: Let $\{\alpha, \beta\}$ be a set of linearly independent vectors in $\mathbb{E}_n$. Let ξ vary such that

$$(\xi - \alpha, \xi - \beta) = 0 \quad (6)$$

Then ξ describes a sphere whose center is defined by $\frac{\alpha+\beta}{2}$ and whose radius is $\left\| \frac{\alpha-\beta}{2} \right\|$.

Proof: One observes that (6) can be written as

$$\left(\left[\xi - \frac{\alpha + \beta}{2} \right] + \frac{\beta - \alpha}{2}, \left[\xi - \frac{\alpha + \beta}{2} \right] + \frac{\alpha - \beta}{2} \right) = 0 \quad (7)$$

One also observes that (6) implies $(\xi - \beta, \xi - \alpha) = 0$, which can be written as

$$\left(\left[\xi - \frac{\alpha + \beta}{2} \right] + \frac{\alpha - \beta}{2}, \left[\xi - \frac{\alpha + \beta}{2} \right] + \frac{\beta - \alpha}{2} \right) = 0 \quad (8)$$

From (7) and (8), we obtain

$$\left\| \xi - \frac{\alpha + \beta}{2} \right\| = \left\| \frac{\alpha - \beta}{2} \right\| \quad (9)$$

Theorem 2.1: Let $\{A, B\}$ be a set of linearly independent transformations on $\mathbb{E}_n$, and X , a linear transformation on $\mathbb{E}_n$, vary such that

$$[X - A]^*[X - B] = 0 \quad (10)$$

Then

$$\left\| X - \frac{A + B}{2} \right\|_H = \left\| \frac{A - B}{2} \right\|_H$$

Proof: Let $\xi \in \mathbb{E}_n$ and $\|\xi\| = 1$. Then equation (10) implies that

$$([X - A]^*[X - B]\xi, \xi) = ([X - B]\xi, [X - A]\xi) = 0$$

or $(X\xi - B\xi, X\xi - A\xi) = 0$. One observes that the vectors $X\xi$, $A\xi$, and $B\xi$ satisfy (6). Thus by (9), we obtain

$$\left\| \left[X - \frac{A + B}{2} \right] \xi \right\| = \left\| \left[\frac{A - B}{2} \right] \xi \right\| \text{ for all } \|\xi\| = 1$$

Consequently, by Theorem 1.3, we have

$$\left\| X - \frac{A + B}{2} \right\|_H = \left\| \frac{A - B}{2} \right\|_H$$

One may employ the Frobenius inner product and norm, and state and prove a theorem analogous to Theorem 2.1. We leave this to the reader as an exercise.

One may also ask for a relation among the singular values of $X - A$, $X - B$, A , and B when the Frobenius inner product is used.

Lemma 2.2: Let $\{\alpha_1, \dots, \alpha_n\}$ be a set of linearly independent vectors in $\mathbb{E}_n$, and let $\xi \in \mathbb{E}_n$ vary such that

$$\sum_{i < j} a_{ij} (\xi - \alpha_i, \xi - \alpha_j) = 0 \quad (11)$$

Then ξ describes a sphere.

One observes that (11) is equivalent to

$$\|\xi\|^2 \left(\sum_{i < j} a_{ij} \right) - \sum_{i < j} a_{ij} [(\xi, \alpha_j) + (\alpha_j, \xi)] + \sum_{i < j} a_{ij} (\alpha_i, \alpha_j) = 0$$

If $\sum a_{ij} = 0$, then (11) describes a hyperplane. Let $\sum a_{ij} \neq 0$. Then one can easily find the center and the radius of the sphere. We shall leave this as an exercise.

A particular case of (11) is when $\sum_{i < j} a_{ij} = 1$. One may also consider the following corollary.

Lemma 2.3: Let $\{\alpha_1, \dots, \alpha_n\}$ be a set of linearly independent vectors in $\mathbb{E}_n$, and let ξ vary such that

$$(\xi - \alpha_i, \xi - \alpha_j) = 0, \quad i \neq j \quad (12)$$

Then ξ describes a sphere with center $\frac{\sum \alpha_i}{n}$ and radius

$$\frac{1}{n\sqrt{n-1}} \left[\sum \|\alpha_i - \alpha_j\|^2 \right]$$

Proof: Expanding (12), we obtain

$$\left\| \xi - \frac{\sum \alpha_i}{n} \right\|^2 = \frac{1}{n^2(n-1)} \left[\sum \|\alpha_i - \alpha_j\|^2 \right]$$

We leave the details to the reader.

Theorem 2.2: Let $\{A_1, \dots, A_n\}$ be a set of linearly independent transformations on $\mathbb{E}_n$, and X be a linear transformation on $\mathbb{E}_n$ varying such that

$$[X - A_i]^* [X - A_j] = 0, \quad i \neq j \quad (13)$$

Then

$$\left\| X - \frac{\sum A_i}{n} \right\|_H^2 = \frac{1}{n^2(n-1)} \left[\sum \|A_i - A_j\|_H^2 \right]$$

Proof: As in Theorem 2.1, we let $\xi \in \mathbb{E}_n$, and $\|\xi\| = 1$. Then (13) implies that

$$(X\xi - A_i\xi, X\xi - A_j\xi) = 0, \quad i \neq j \quad (14)$$

Applying (12) to (14), we obtain

$$\left\| X - \frac{\sum A_i}{n} \right\|_H^2 = \frac{1}{n^2(n-1)} \left[\sum \|A_i - A_j\|^2 \right]$$

The reader may apply (12) to the Frobenius inner product and norm, and state and prove a theorem analogous to Theorem 2.2.

Theorem 2.2 may also be extended to rectangular matrices. We shall leave this to the reader.

Lemma 2.4: Let $\{\alpha, \beta\}$ be a set of linearly independent vectors in $\mathbb{E}_n$, and let ξ vary such that

$$\|\xi - \alpha\|^2 + \|\xi - \beta\|^2 = k^2 \quad (15)$$

Then ξ describes a sphere whose center is $\frac{1}{2}(\alpha + \beta)$ and whose radius depends on k .

Proof: From (15), we obtain

$$2\|\xi\|^2 - (\xi, \alpha + \beta) - (\alpha + \beta, \xi) + \|\alpha\|^2 + \|\beta\|^2 = k^2$$

which implies

$$\left\| \xi - \frac{\alpha + \beta}{2} \right\|^2 - \left\| \frac{\alpha + \beta}{2} \right\|^2 = \frac{1}{2} [k^2 - \|\alpha\|^2 - \|\beta\|^2]$$

It is clear that the above equation represents a sphere whose center is $\frac{\alpha + \beta}{2}$. Now in order to have a real number for the radius, we must have

$$\frac{1}{2} [k^2 - \|\alpha\|^2 - \|\beta\|^2] + \left\| \frac{\alpha + \beta}{2} \right\|^2 > 0$$

or $k^2 > \frac{1}{2}\|\alpha - \beta\|^2$. We leave the details to the reader.

Theorem 2.3: Let $\{A, B\}$ be a set of linearly independent transformations on $\mathbb{E}_n$, and let X vary such that

$$\|X - A\|_F^2 + \|X - B\|_F^2 = k^2$$

Then

$$\left\| X - \frac{A + B}{2} \right\|_F = \text{constant}$$

Proof: Considering the quadratic forms of $(X - A)^*(X - A)$ and $(X - B)^*(X - B)$ for $\|\xi\| = 1$, and Theorem 1.1, one finds that the constant is $\frac{k^2}{2} - \left\| \frac{A - B}{2} \right\|_F^2$. One must choose $k^2 \geq \frac{1}{2}\|A - B\|_F^2$; otherwise, the constant will be imaginary. We leave the details to the reader.

The reader may apply Lemma 2.4 to obtain a similar theorem for the Hilbert norm.

This theorem can also be generalized to obtain inequalities involving singular values of $X - A$ and $X - B$.

Lemma 2.5: Let $\{\alpha_1, \dots, \alpha_n\}$ be linearly independent in $\mathbb{E}_n$, and let ξ vary such that

$$\sum_{i=1}^n \|\xi - \alpha_i\|^2 = k^2 \quad (16)$$

Then ξ describes a sphere whose center is $\frac{1}{n} \sum_{i=1}^n \alpha_i$, and whose radius depends on k .

Proof: We shall give an outline of a proof. From (16), we obtain

$$\|\xi\|^2 - \left\{ \left(\xi, \frac{\sum \alpha_i}{n} \right) + \left(\frac{\sum \alpha_i}{n}, \xi \right) \right\} + \frac{1}{n} \sum \|\alpha_i\|^2 = \frac{k^2}{n}$$

This implies that

$$\left\| \xi - \frac{\sum \alpha_i}{n} \right\|^2 - \left\| \frac{\sum \alpha_i}{n} \right\|^2 + \frac{1}{n} \sum \|\alpha_i\|^2 = \frac{k^2}{n}$$

One observes that the radius is

$$r = \frac{k^2}{n} - \frac{1}{n} \sum \|\alpha_i\|^2 + \left\| \frac{\sum \alpha_i}{n} \right\|^2$$

Conjecture 2.1: Let A_i $i = 1, \dots, n$ be linearly independent on $\mathbb{E}_n$, and let X vary so that

$$\sum_{i=1}^n \|X - A_i\|_H^2 = k^2$$

Then

$$\left\| X - \frac{\sum A_i}{n} \right\|_H = p \quad (17)$$

is constant, and

$$p = \frac{k^2}{n} - \frac{1}{n} \sum \|A_i\|_H^2 + \left\| \frac{\sum A_i}{n} \right\|_H^2$$

Suggestion: Considering the sum of quadratic forms of $(X - A_i)^*(X - A_i)$ for $\|\xi\| = 1$, and applying 2.5 and 1.1, one may obtain (17) or an inequality. We leave the details to the reader.

3 Suggestions

1. In Lemma 2.1, one may study the case when $(\xi - \alpha, \xi - \beta) = c$, a constant, and attempt to obtain theorems similar to 2.1.

2. In Lemma 2.3, one may study the cases when

$$(\xi\alpha_i, \xi\alpha_j) = c, \quad i \neq j \quad (18)$$

and

$$(\xi - \alpha_i, \xi - \alpha_j) = c_{ij}, \quad i \neq j \quad (19)$$

Note that in (18), c is the same for all i and j , but in (19) each equality has a different constant.

3. In Lemma 2.4, one may replace (15) by $||\xi - \alpha||^2 - ||\xi - \beta||^2 = k^2$, and try to apply it to the norms of some linear transformations.

4. Lemma 2.5 can be changed by setting $\left| \sum_{i=1}^j ||\xi - \alpha_i||^2 - \sum_{i=j+1}^n ||\xi - \alpha_i||^2 \right| = k^2$, and considering theorems similar to 2.1.

5. One may study $\frac{||\xi - \alpha||}{||\xi - \beta||} = k > 0$, its generalizations, and applications to linear transformations.

In Conjecture 2.1, a sum of quadratic forms is considered. One may attempt to obtain many inequalities using the *minimax principle* [1].

References

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2. Ali R. Amir-Moéz, *Elements of Geometry in Unitary Spaces*, Monographs in Undergraduate Math., Vol. 7.